

Same translation, different density widths

$$\phi_\mu(x) = x + \mu, \quad \mu = 0, \quad d\mu = 0.6 \quad | \quad \text{Dashed density and arrows: } \varepsilon d\mu = 0.3$$

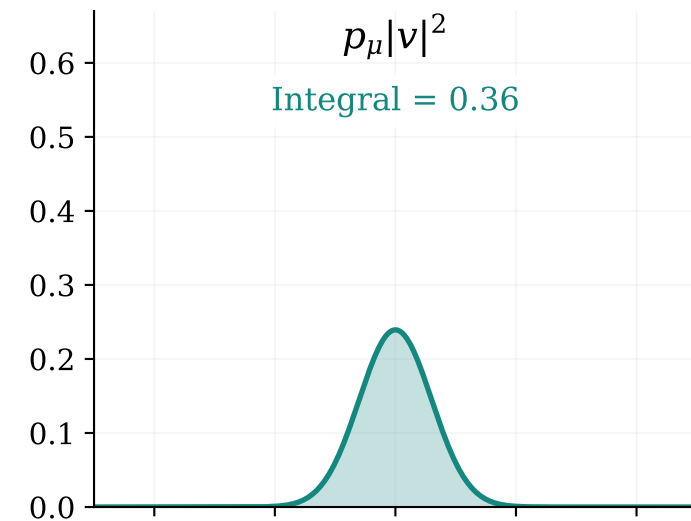
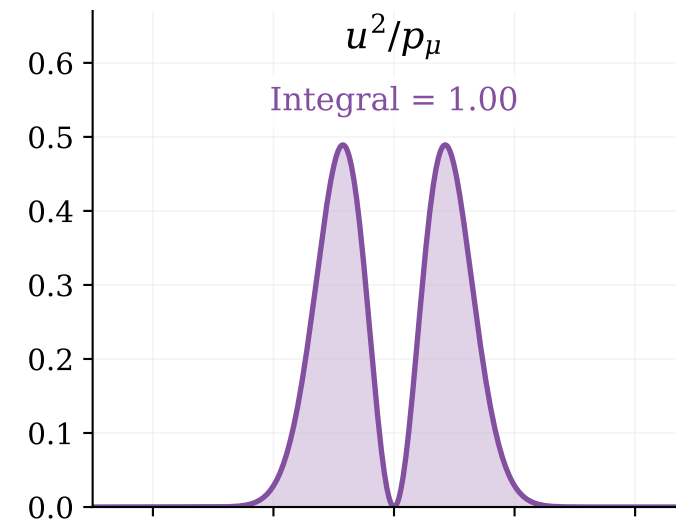
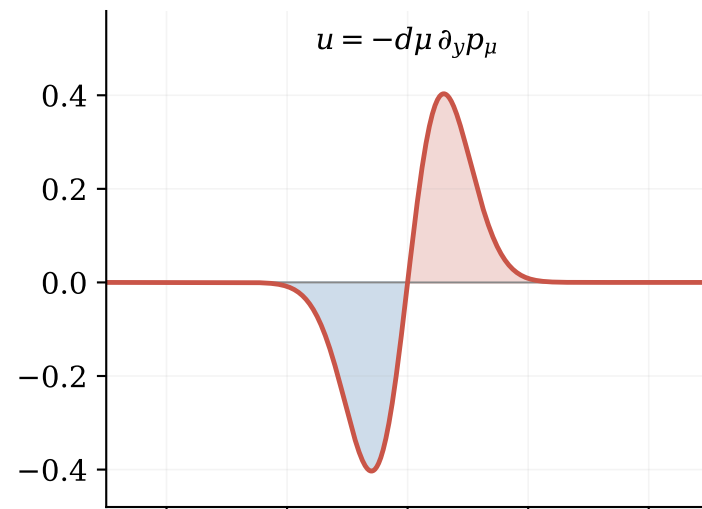
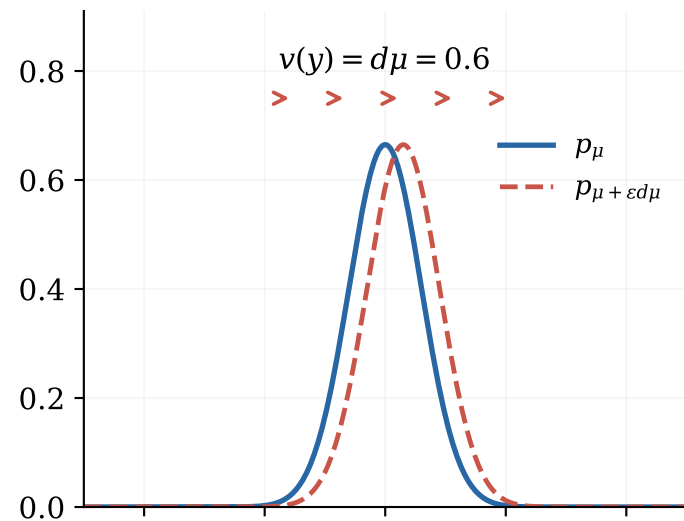
A constant translation

Density variation

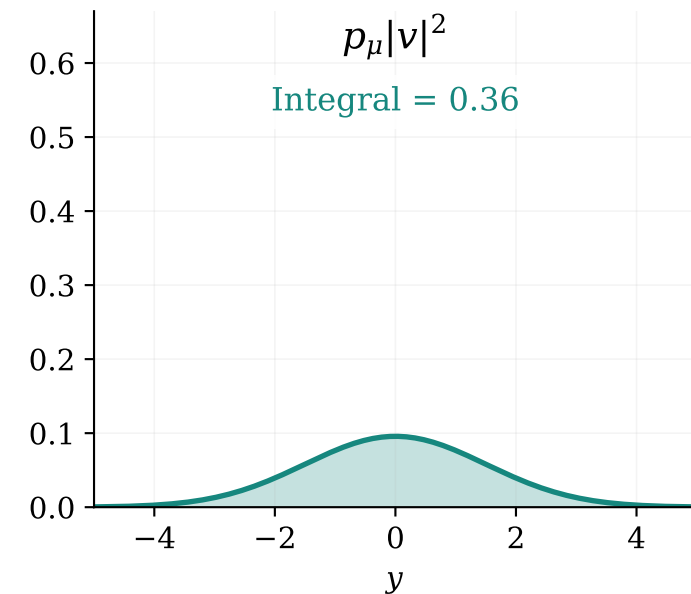
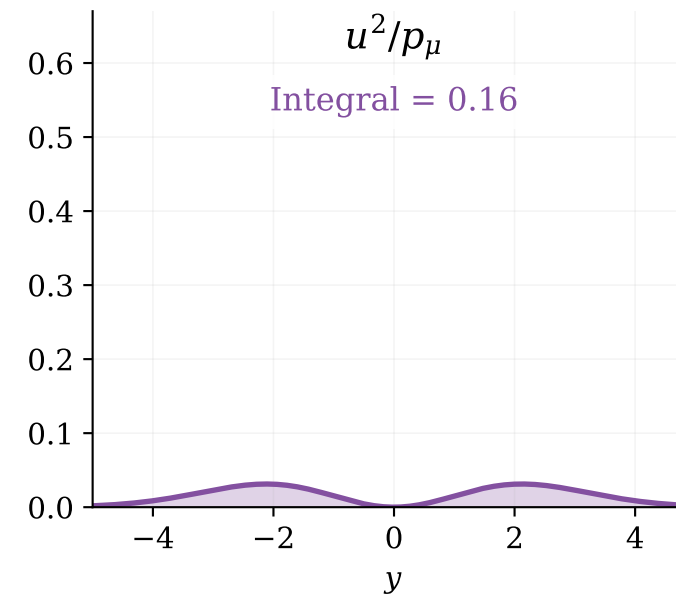
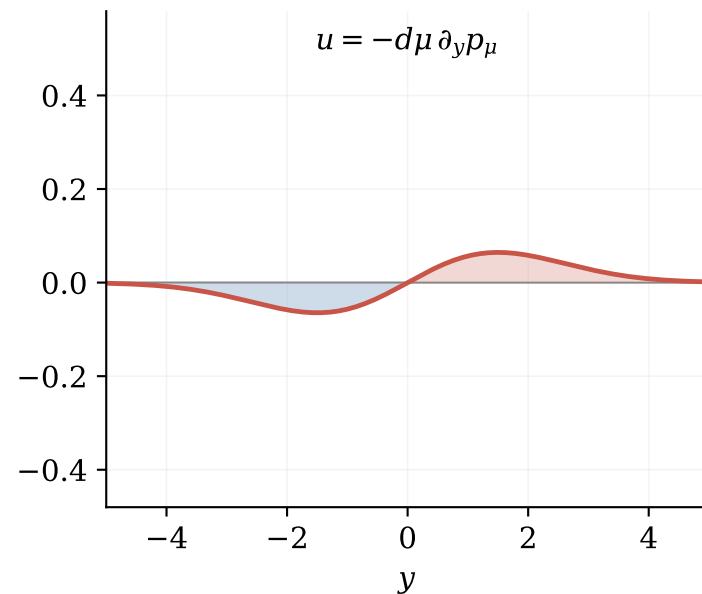
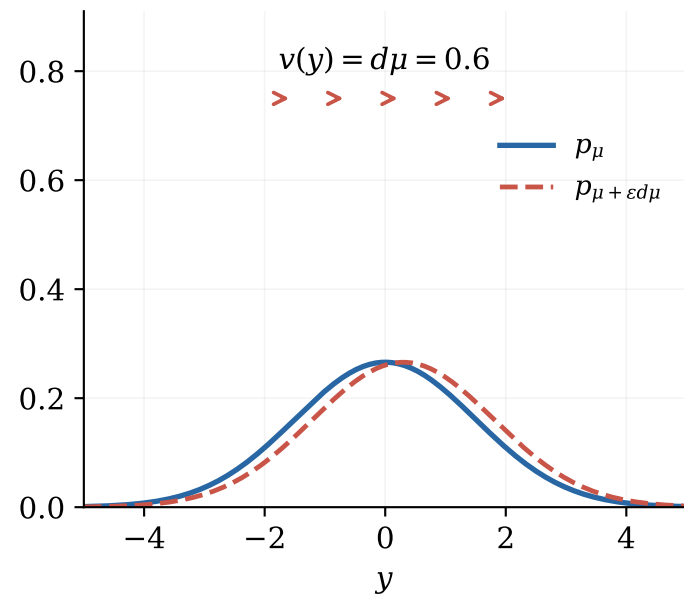
Fisher-Rao

Wasserstein-Otto

Narrow
 $\sigma = 0.6$



Broad
 $\sigma = 1.5$



FR: $(d\mu)^2/\sigma^2$ — sensitive to width

WO: $(d\mu)^2$ — cost of displacement